EXPERIMENT 7: HOUSE PRICE PREDICTION AND CLUSTERING

AIM:

To analyze the housing dataset, handle missing values, explore relationships among housing features, build and evaluate Linear Regression, Polynomial Regression, and Support Vector Regression models to predict median house value, and apply K-Means clustering to group housing records.

ALGORITHM:

1. Import the required libraries, including Pandas, NumPy, Seaborn, Matplotlib, train_test_split, LinearRegression, PolynomialFeatures, make_pipeline, SVR, evaluation metrics, and KMeans.

2. Load the housing CSV dataset into a Pandas DataFrame and display the first five records.

3. Inspect the dataset information, statistical description, data types, and missing values.

4. Select housing_median_age, total_rooms, total_bedrooms, and median_income as input features (X), and median_house_value as the target (y).

5. Visualize the selected features and their relationships using a pair plot.

6. Fill missing values in the selected input features using forward fill.

7. Split the dataset into training and testing sets using train_test_split with test_size=0.2 and random_state=42.

8. Train a Linear Regression model and generate predictions for the test data.

9. Train a Polynomial Regression model using PolynomialFeatures with degree 2 and generate predictions.

10. Train a Support Vector Regression model using an RBF kernel and generate predictions.

11. Evaluate each model using Mean Absolute Error (MAE) and Mean Squared Error (MSE), as shown in the output.

12. Apply K-Means clustering with three clusters to the housing data and add the cluster labels to the DataFrame.

13. Visualize the house-price clusters using a scatter plot of total_rooms against median_house_value.

```python
# EXERCISE-5 MUTHURAJ (241801176)
# Step 1: Import Required Libraries

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.pipeline import make_pipeline
from sklearn.svm import SVR
from sklearn.metrics import mean_absolute_error, mean_squared_error
from sklearn.cluster import KMeans
```

```python
# EXERCISE-5 MUTHURAJ (241801176)
# Step 2: Load Dataset

df = pd.read_csv(r"C:\\Users\\ADMIN\\Downloads\\archive (3)\\housing.csv")

print("First 5 rows:")
print(df.head())
```

```
First 5 rows:
   longitude  latitude  housing_median_age  total_rooms  total_bedrooms  \
0    -122.23     37.88                41.0        880.0           129.0
1    -122.22     37.86                21.0       7099.0          1106.0
2    -122.24     37.85                52.0       1467.0           190.0
3    -122.25     37.85                52.0       1274.0           235.0
4    -122.25     37.85                52.0       1627.0           280.0

   population  households  median_income  median_house_value ocean_proximity
0       322.0       126.0         8.3252            452600.0        NEAR BAY
1      2401.0      1138.0         8.3014            358500.0        NEAR BAY
2       496.0       177.0         7.2574            352100.0        NEAR BAY
3       558.0       219.0         5.6431            341300.0        NEAR BAY
4       565.0       259.0         3.8462            342200.0        NEAR BAY
```

```python
# EXERCISE-5 MUTHURAJ (241801176)
# Step 3: Explore Dataset

print("\nDataset Information:")
print(df.info())

print("\nStatistical Description:")
print(df.describe())
```

```
Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 10 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   longitude           20640 non-null  float64
 1   latitude            20640 non-null  float64
 2   housing_median_age  20640 non-null  float64
 3   total_rooms         20640 non-null  float64
 4   total_bedrooms      20433 non-null  float64
 5   population          20640 non-null  float64
 6   households          20640 non-null  float64
 7   median_income       20640 non-null  float64
 8   median_house_value  20640 non-null  float64
 9   ocean_proximity     20640 non-null  str
dtypes: float64(9), str(1)
memory usage: 1.7 MB
None

Statistical Description:
          longitude      latitude  housing_median_age   total_rooms  \
count  20640.000000  20640.000000        20640.000000  20640.000000
mean    -119.569704     35.631861           28.639486   2635.763081
std        2.003532      2.135952           12.585558   2181.615252
min     -124.350000     32.540000            1.000000      2.000000
25%     -121.800000     33.930000           18.000000   1447.750000
50%     -118.490000     34.260000           29.000000   2127.000000
75%     -118.010000     37.710000           37.000000   3148.000000
```

```python
# EXERCISE-5 MUTHURAJ (241801176)
# Step 4: Check Missing Values

print("\nMissing Values:")
print(df.isnull().sum())
```

```
Missing Values:
longitude             0
latitude              0
housing_median_age    0
total_rooms           0
total_bedrooms      207
population            0
households            0
median_income         0
median_house_value    0
ocean_proximity       0
dtype: int64
```

```python
# EXERCISE-5 MUTHURAJ (241801176)
# Step 5: Select Features and Target

X = df[['housing_median_age', 'total_rooms', 'total_bedrooms', 'median_income']]
y = df['median_house_value']

print("\nFeatures:")
print(X.head())

print("\nTarget:")
print(y.head())
```

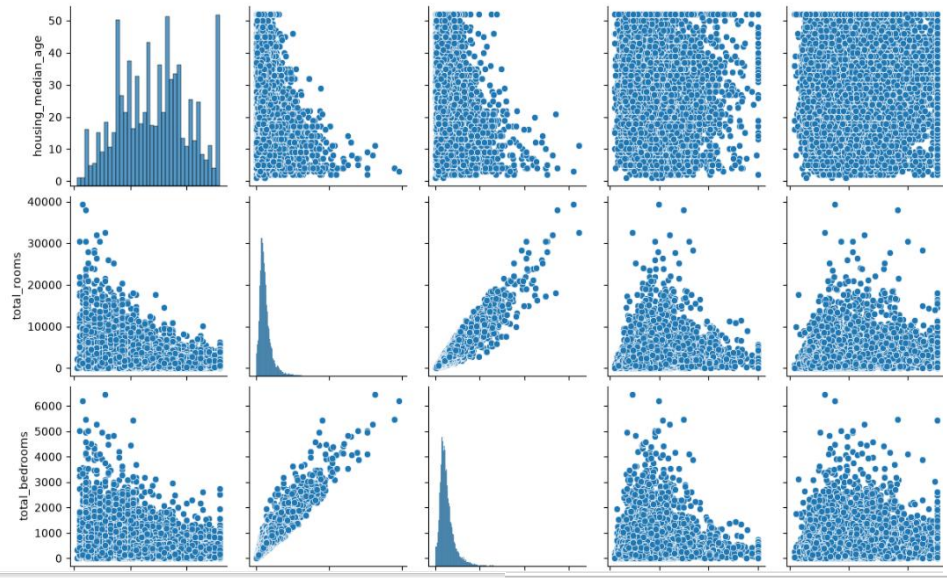
```
Features:
   housing_median_age  total_rooms  total_bedrooms  median_income
0                41.0        880.0           129.0         8.3252
1                21.0       7099.0          1106.0         8.3014
2                52.0       1467.0           190.0         7.2574
3                52.0       1274.0           235.0         5.6431
4                52.0       1627.0           280.0         3.8462

Target:
0    452600.0
1    358500.0
2    352100.0
3    341300.0
4    342200.0
Name: median_house_value, dtype: float64
```

```
# EXERCISE-5 MUTHURAJ (241801176)
# Step 6: Visualize Relationships

sns.pairplot(df[['housing_median_age', 'total_rooms',
                 'total_bedrooms', 'median_income',
                 'median_house_value']])

plt.show()
```




```
# EXERCISE-5 MUTHURAJ (241801176)
# Step 6.5: Handle Missing Values

X = X.fillna(X.median())

print("Missing values:")
print(X.isnull().sum())
```

```
Missing values:
housing_median_age    0
total_rooms           0
total_bedrooms        0
median_income         0
dtype: int64
```

```
# EXERCISE-5 MUTHURAJ (241801176)
# Step 7: Split Dataset

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
# EXERCISE-5 MUTHURAJ (241801176)
# Step 8: Linear Regression

linear_model = LinearRegression()

linear_model.fit(X_train, y_train)

y_pred_linear = linear_model.predict(X_test)
```

```
# EXERCISE-5 MUTHURAJ (241801176)
# Step 9: Polynomial Regression

polynomial_model = make_pipeline(
    PolynomialFeatures(degree=2),
    LinearRegression()
)

polynomial_model.fit(X_train, y_train)
y_pred_poly = polynomial_model.predict(X_test)

print("Polynomial Regression completed successfully!")
```

```
Polynomial Regression completed successfully!
```

```
# EXERCISE-5 SYED MUTHURAJ (241801176)
# Step 10: Support Vector Regression

svr_model = SVR(kernel='rbf')

svr_model.fit(X_train, y_train)
y_pred_svr = svr_model.predict(X_test)

print("SVR completed successfully!")
```

```
SVR completed successfully!
```

```python
# EXERCISE-5MUTHURAJ (241801176)
# Step 11: Model Evaluation

models = {
    "Linear Regression": y_pred_linear,
    "Polynomial Regression": y_pred_poly,
    "SVR": y_pred_svr
}

for name, prediction in models.items():
    mae = mean_absolute_error(y_test, prediction)
    mse = mean_squared_error(y_test, prediction)
    print(name)
    print("MAE:", mae)
    print("MSE:", mse)
    print()
```

```
Linear Regression
MAE: 58342.91635674605
MSE: 6356567901.161352

Polynomial Regression
MAE: 62252.22762860845
MSE: 7834004738.55757

SVR
MAE: 87333.38074091132
MSE: 13739650709.806314
```

```python
# EXERCISE-5 MUTHURAJ (241801176)
# Step 12: K-Means Clustering

kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
kmeans.fit(X)

df["Cluster"] = kmeans.labels_

print(df[["Cluster"]].head())
```

```
   Cluster
0        1
1        0
2        1
3        1
4        1
```

```python
# EXERCISE-5 MUTHURAJ (241801176)
# Step 13: Visualize Clusters

sns.scatterplot(
    data=df,
    x="total_rooms",
    y="median_house_value",
    hue="Cluster"
)

plt.title("House Price Clusters")
plt.show()
```



RESULT:

The housing dataset was loaded and explored, and missing values were checked and handled in the selected features. Linear Regression, Polynomial Regression, and Support Vector Regression models were trained to predict median house value and evaluated using MAE and MSE. K-Means clustering with three clusters was applied, and the house-price clusters were visualized using total rooms and median house value.